

Md. Manuaruzzaman

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SUMMARY

Self-motivated Machine Learning engineer, with two years of experience on ML & Computer Vision achieved through diverse projects including various data processing, analysis, visualization, predictive modeling and performance optimization. An Engineering graduate passionate about artificial intelligence and eager to contribute to real-world applications, A self-learner with data driven analytical mindset, continuously seeking skill enhancement and following industry trends.

EXPERIENCE

Innovative Skills Ltd.

Full-Stack AI Engineer – Full-time

Oct 2024 – present

Research, develop and integrate state of the art computer vision models as part of embedded system and software services. Time Series Analysis, Sentiment Analysis for business growth, service automation.

Instructor – Image Processing for Computer Vision

Sep 2024 – present

Freelancer – ML Engineer

March 2023 – Oct 2024

eShikhon.com – Machine Learning Instructor

Oct 2024 – present

TechKnowGram Ltd. – AI Engineer Intern (Contractual)

Jul – Sep 2024

Incorporated state of the art computer vision model in webapp for achieving better result. Developed a service management software with role-based access control and task monitoring option.

School of Future Digital Lab – Python Trainer (Contractual)

Feb – Jun 2024

Freelancing – Web Development

Aug 2020 – Aug 2021

SKILLS

- **Programming Language:** Python, R
- **Technical Skills:** CNN, Deep Learning, Image Processing & Segmentation, Data Augmentation, Object Detection, Face Detection & Recognition
- **Version Control:** Git, GitHub
- **OS & Tools:** Linux, Windows 10
- **Libraries:** OpenCV, Matplotlib, TensorFlow, Keras, Scikit-Learn, NumPy, Pandas, SciPy, Seaborn
- **Soft Skills:** Critical thinking, Problem Solving, Time management, Go-getter attitude, Collaboration, Communication, Desire to learn, Decision Making
- **Language:** Bengali (Native), English (Proficient)

PROJECTS

Self-Supervised Deblurring Model using - Computer Vision - [Explore](#)

A self-supervised DnCNN based deblurring model was built and trained on CamVid Dataset from Kaggle

CamVid Semantic Segmentation Model - Computer Vision - [Explore](#)

An image segmentation model was built based on FCN-8 model and trained on CamVid Dataset from Kaggle. It segments images in different regions according to objects and masks the regions according to their labeled color.

Technologies: Image Processing, Semantic Segmentation, U-Net, FCN-8, CNN

Resume Classifier - Machine Learning - [Explore](#)

This ML model classifies resumes into 24 categories. A Kaggle dataset was used to train the model. Developed a python script, which runs on the command line.

Technologies: NLP, NLTK, TF-IDF Vectorizer, Joblib, Pipeline

Telco Churn Prediction - ML Deployment - [Explore](#) – [Try It](#)

A Web App utilizing DjangoRestFramework integrated with a binary classification model customer churn prediction.

Technologies: Django, DjangoRestFramework, Scikit-Learn, NumPy, Pandas, Matplotlib

THESIS WORKS

Diabetic retinopathy classification from fundus images using Convolutional Neural Network (CNN) - Deep Learning

A simple 5-layer CNN model, trained using APTOS 2019 blindness detection dataset from Kaggle. Achieved an excellent accuracy of 96% and f1-score of 96%.

Lower Backpain Identification - Machine Learning

A binary classification model trained with a very small dataset from Kaggle, utilized different Feature Selection methods, Hyperparameter tuning and ensemble learning to achieve an accuracy of 91.94%.

EDUCATION

Hajee Mohammad Danesh Science & Technology University - Dinajpur, Bangladesh

B.Sc. (Engineering) in Electrical and Electronic Engineering